

File 4: Sub-Phase 2.4 — Technical Stack

PHASE 2: REQUIREMENT ANALYSIS

Sub-Phase 2.4: Technical Stack Identification

Field	Details
Date	28-09-2026
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Project Name	Implement Client Script & UI Policy (Incident)
Document Type	Sub-Phase Deliverable 2.4 — Technical Stack Identification

1. Overview & Architecture Stack

The Technical Stack defines the core ServiceNow platform technologies, frameworks, APIs, and client-side execution components required to build, test, and deploy the solution on the incident table.

2. Core Technical Stack Breakdown

Layer / Domain	Technology Component	Version / Specification	Role in Implementation
Platform Instance	ServiceNow Cloud Platform	Washington DC / Xanadu Release	Enterprise cloud infrastructure hosting the ITSM application.
Target Data Model	Incident Table (incident)	Core ITSM Data Structure	Target database table containing fields (impact, urgency, assignment_group, assigned_to, state).
Declarative Engine	ServiceNow UI Policy Engine	Native Platform Framework	Controls dynamic field states (Mandatory, Read-Only) client-side without code.
Scripting Framework	ServiceNow Client Scripts	ECMAScript 5/6 JavaScript	Event-driven client scripts (onChange, onSubmit, onCellEdit).
Client API	GlideForm (g_form) API	ServiceNow Client-Side API	Provides UI functions (getValue, setValue, showErrorMessage, addInfoMessage).
Deployment Mechanism	ServiceNow Update Sets	Platform Utility	Captures and migrates system configurations across sub-production instances.

3. API Methods & Function Scope

Client API Function	Function Scope & Purpose	Target Field Usage
<code>g_form.getValue(fieldName)</code>	Retrieves real-time value from a form field.	Checked in <code>onSubmit</code> to read impact and <code>assigned_to</code> .
<code>g_form.setValue(fieldName, value)</code>	Programmatically updates a field's value.	Used in <code>onChange</code> to auto-set urgency to 1.
<code>g_form.addInfoMessage(message)</code>	Displays a blue informational banner at the top of the form.	Used in <code>onChange</code> to notify users of auto-updated urgency.
<code>g_form.showErrorBox(fieldName, message)</code>	Displays a targeted red error box beneath a specific field.	Used in <code>onSubmit</code> to highlight missing <code>assigned_to</code> .
<code>callback(false)</code>	Intercepts and cancels inline list cell editing.	Used in <code>onCellEdit</code> to prevent editing state from list vie